The documentation and process conversion measures necessary to comply with this revision shall be completed by 6 January 1994

INCH-POUND

MIL-S-19500/348D
6 October 1993
SUPERSEDING
MIL-S-19500/348C(EC)
26 May 1987

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING TYPE 2N3467, 2N3467L, 2N3468, 2N3468L, JANTX, JANTXV, AND JANS

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNP silicon switching transistors. Three levels of product assurance is provided for the device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3 (similar to TO-5 and TO-39).

1.3 Maximum ratings.

P_T 1/	P_T 2/	V_{CBO}		V_{CEO}		V_{EBO}	I_C	T_{OP} and T_{STG}
T_A = +25°C	T_C = +25°C	2N3467 2N3467L	2N3468 2N3468L	2N3467 2N3467L	2N3468 2N3468L			
W	W	V dc	V dc	V dc	V dc	V dc	A dc	°C
1.0	5.0	40	50	40	50	5.0	1.0	-55 to +175

1/ Derate linearly at 5.71 mW/°C above T_A = +25°C.
2/ Derate linearly at 28.6 mW/°C above T_C = +25°C.

1.4 Primary electrical characteristics.

	h_{FE2} 1/		h_{FE3} 1/		$V_{CE(sat)}$
LIMIT	V_{CE} = 1.0 V dc I_C = 500 mA dc	V_{CE} = 1.0 V dc I_C = 500 mA dc	V_{CE} = 5.0 V dc I_C = 1.0 A dc	V_{CE} = 5.0 V dc I_C = 1.0 A dc	I_C = 500 mA dc I_B = 50 mA dc
	2N3467 2N3467L	2N3468 2N3468L	2N3467 2N3467L	2N3468 2N3468L	
					V dc
MIN	40	25	40	25	
MAX	120	75			0.6

	C_{obo}	f_T		t_{on}	t_{off}
LIMITS	V_{CB} = 10 V dc I_E = 0; 100 kHz ≤ f ≤ 1 MHz	V_{CE} = 10 V dc I_C = 50 mA dc f = 100 MHz		I_C = 500 mA dc I_B = 50 mA dc	I_C = 500 mA dc I_B = 50 mA dc
		2N3467 2N3467L	2N3468 2N3468L		
	pF	MHz	MHz	ns	ns
MIN		175	150		
MAX	25	500	500	40	90

1/ Pulsed (see 4.5.1)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (see 6.2)

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 Appendix F, Figure 8.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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4.3 Screening (JANTX, JANTXV, and JANS only.) Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{CBO1} and h_{FE2}	Not applicable
11	I_{CBO1}; h_{FE2}; ΔI_{CBO1} = 100% of initial value or 50 nA dc, whichever is greater; Δh_{FE2} = 25% of initial value.	I_{CBO1} and h_{FE2}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 & 3 of table I herein; ΔI_{CBO1} = 100% of initial value or 50 nA dc, whichever is greater; Δh_{FE2} = 25% of initial value.	Subgroup 2 of table I herein; ΔI_{CBO1} = 100% of initial value or 50 nA dc, whichever is greater; Δh_{FE2} = 25% of change initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:
T_A = Room ambient as defined in the general requirements of MIL-STD-750, paragraph 4.5; P_T = 1.0 W;
V_{CB} = 30 V dc for 2N3467
V_{CB} = 40 V dc for 2N3468
V_{CB} = 10 V dc for JANS devices.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
3	2037	Test condition A.
4	1037	V_{CB} = 10 V dc; P_T = 1.0 W at T_A = +25° ±3°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted.
5	1027	V_{CB} = 10 V dc; T_A = +125°C ±25°C for 96 hours, P_T = 1.0 W at T_A = +100°C or adjusted as required according to the chosen T_A to give an average T_J = +275°C.
5	2037	Test condition A.
6	3131	See 4.5.3.

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4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
3	1027	$V_{CB} \geq 10$ V dc; $P_T = 1.0$ W at $T_A = +30°\pm5°C$. No heat sink or forced-air cooling on the devices shall be permitted.
3	2037	Test condition A.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
2	2036	Test condition E.
6	1026	$V_{CB} \geq 10$ V dc; $P_T = 1.0$ W at $T_A = +30° \pm5°C$. No heat sink or forced-air cooling on the device shall be permitted.

4.4 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.4.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.4.2 Input capacitance. This test shall be conducted in accordance with MIL-STD-750, method 3240, except the output capacitor shall be omitted.

4.4.3 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. the following details shall apply:

a. Minimum collector current magnitude during power application shall be 160 mA dc for $R_{\theta JC}$ and 32 mA dc for $R_{\theta JA}$.

b. Collector to base voltage magnitude shall be 10 V dc.

c. Reference temperature measuring point shall be the case for $R_{\theta JC}$ and ambient air for $R_{\theta JA}$.

d. Reference point temperature shall be selected with $25°C \leq T_R \leq 35°C$ and recorded before test is started.

e. Mounting arrangement shall be with heat sink to case for $R_{\theta JC}$ and without heat sink for $R_{\theta JA}$.

f. Maximum limits shall be $R_{\theta JC} = 35°C/W$ and $R_{\theta JA} = 175°C/W$.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage collector to base	3001	Bias condition D; I_C = 10 µA dc	$V_{(BR)CBO}$			
2N3467				40		V dc
2N3468				50		V dc
Breakdown voltage emitter to base	3026	Bias condition D; I_E = 10 µA dc	$V_{(BR)EBO}$	5.0		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N3467				40		V dc
2N3468				50		V dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO1}		100	nA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{EB} = 3.0 V dc; V_{CE} = 30 V dc	I_{CEX}		100	nA dc
Forward to current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE1}			
2N3467				40		
2N3468				25		
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE2}			
2N3467				40	120	
2N3468				25	75	
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)	h_{FE3}			
2N3467				40		
2N3468				25		
Collector to emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(SAT)1}$		0.35	V dc
Collector to emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(SAT)2}$		0.6	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to emitter saturation voltage	3071	I_C = 1.0 A dc; I_B = 100 mA dc; pulsed (see 4.5.1)	$V_{CE(SAT)3}$		1.2	V dc
Base to emitter voltage saturated	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(SAT)1}$		1.0	V dc
Base to emitter voltage saturated	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(SAT)2}$	0.8	1.2	V dc
Base to emitter voltage saturated	3066	Test condition A; I_C = 1.0 A dc; I_B = 100 mA dc; pulsed (see 4.5.1)	$V_{BE(SAT)3}$		1.6	V dc
Subgroup 3						
High-temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO2}		50	µA dc
Low-temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE4}			
2N3467				16		
2N3468				10		
Subgroup 4						
Extrapolated unity gain frequency	3261	V_{CE} = 10 V dc; I_C = 50 mA dc; f = 100 MHz	f_t			
2N3467				175	500	MHz
2N3468				150	500	MHz
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		25	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz (see 4.5.2)	C_{ibo}		100	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Pulse response:						
Delay time	3251	Test condition A; I_C = 500 mA dc; I_{B1} = 50 mA dc; V_{EB} = 2 V dc (see fig. 1)	t_d		10	ns
Rise time	3251	Test condition A; I_C = 500 mA dc; I_{B1} = 50 mA dc; V_{EB} = 2 V dc (see fig.1)	t_r		30	ns
Storage time	3251	Test condition A I_C = 500 mA dc; I_{B1} = I_{B2} = 50 mA dc; (see fig. 2)	t_s		60	ns
Fall Time	3251	Test condition A I_C = 500 mA dc; I_{B1} = I_{B2} = 50 mA dc; (see figure 2)	t_f		30	ns
Subgroups 5, 6, and 7						
Not applicable						

For sampling plan, see MIL-S-19500.

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TABLE II. Groups B and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO1}		100	nA dc
2.	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO1}		200	nA dc
3.	Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(SAT)2}$		0.6	V dc
4.	Base to emitter voltage (saturated)	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(SAT)2}$	0.8	1.2	V dc
5.	Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE2}			
	2N3467				40	120	
	2N3468				25	75	
6.	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	ΔI_{CBO1}	100% of initial value or 50 nA dc, whichever is greater.		
7.	Collector to emitter voltage saturated	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$\Delta V_{CE(SAT)2}$	50 mV dc change from initial value.		
8.	Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	Δh_{FE2}	25% change from initial value.		

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 3, 4, and 5.
b. Subgroup 4, see table II herein, steps 1, 3, 4, 5, and 7.
c. Subgroup 5, see table II herein, steps 1, 3, 4, 5, 6, 7, and 8.

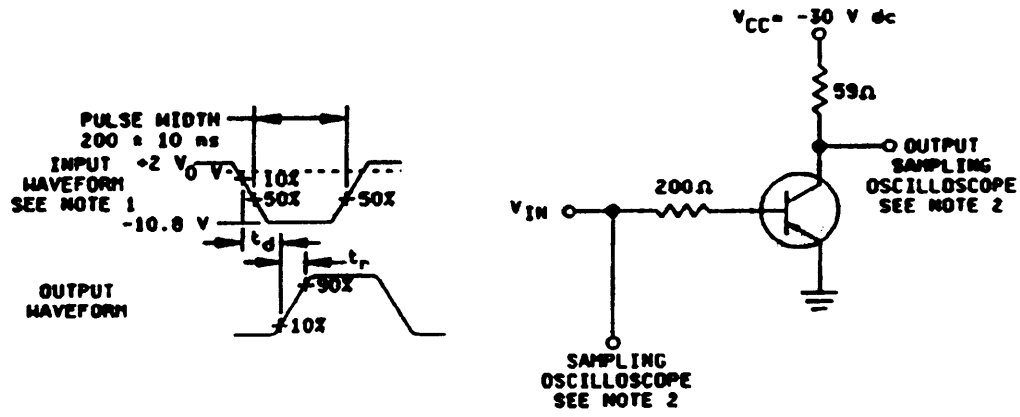
2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 5.
b. Subgroups 3 and 6, see table II herein, steps 2, 5, and 8.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:

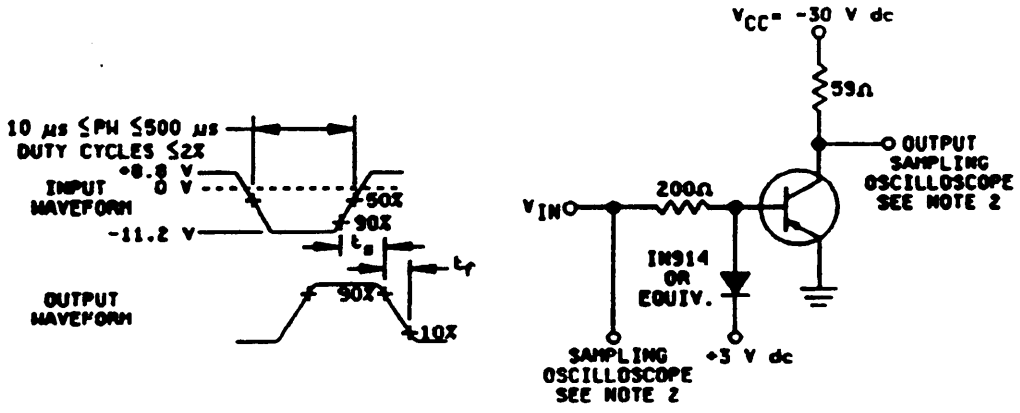
a. Subgroups 2 and 3, see table II herein, steps 1, 3, 4, and 5 (JANS) and 1 and 5 (for JANTX and JANTXV.)
b. Subgroup 6, see table II herein, steps 1, 3, 4, 5, 6, and 8 (for JANS) and 2 and 8 (for JANTX and JANTXV).

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NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2 ns, duty cycle ≤ 2% and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ pF, rise time ≤ .2 ns.

FIGURE 1. Equivalent circuit for measuring delay and rise times.



NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2 ns, duty cycle ≤ 2% and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ pF, rise time ≤ .2 ns.

FIGURE 2. Equivalent circuit for measuring storage and fall times.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1).

b. Lead finish as specified (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, AV, MI
Navy - AS, CG, MC
Air Force - 13, 15, 19, 85
DLA - ES

Preparing activity:
Navy - EC

Agent:
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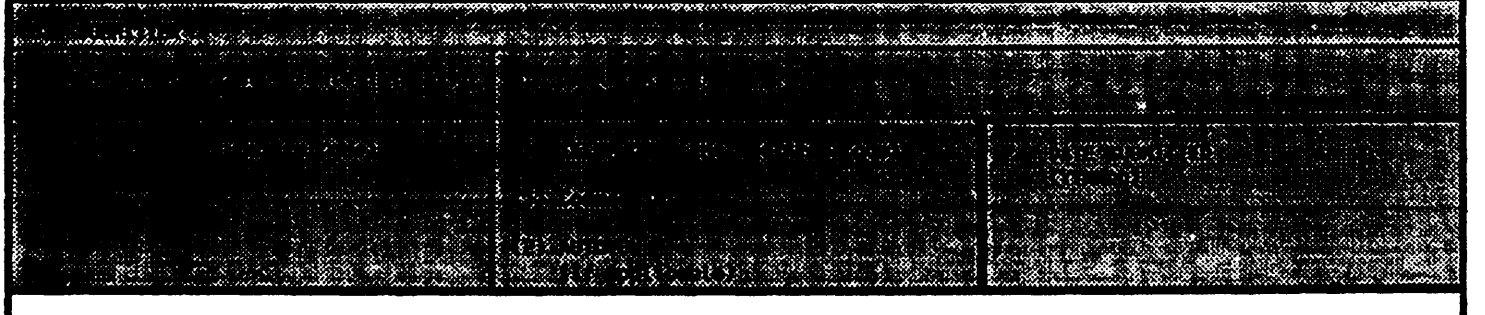
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/368D	2. DOCUMENT DATE (YYMMDD) 93/10/06

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING TYPE 2N3467, 2N3467L, 2N3468, 2N3468L, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION



8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial: 513-296-6048 (2) AUTOVON: 986-6048
c. ADDRESS (Include Zip Code) DESC-ECT Dayton, Ohio 45444-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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